

ORF307: Optimization

Precept: #10

Agenda

- Exercises 3, 6, and 7 from Spring 2023 midterm-2 exam **preparation** (see *AdditionalExercises_02_midterm.pdf* file on Canvas/Modules → "Midterm2 past exams for practice")

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Exercise 3

Let $A = A^T$ be symmetric matrix. Consider the LP

$$\begin{array}{ll} \min_x & c^T x \\ \text{s.t.} & Ax \geq c \\ & x \geq 0 \end{array} \tag{1}$$

Show that if x^* satisfies $Ax^* = c, x^* \geq 0$, then x^* is optimal.

Solution:

Exercise 6

The matrix $P \in \mathbb{R}^{n \times n}$ is a column stochastic if

- $P \geq 0$ (i.e., $p_{ij} \geq 0, \forall i, j$)
- $P^T \mathbf{1} = \mathbf{1}$ (i.e., the column of P sum up to 1, $\sum_{i=1}^n p_{ij} = 1, \forall j$.)

NOTE: Do not confuse the conditions: $P \geq 0$ and $Q \succeq 0$. The former means that all the elements of the matrix P are nonnegative numbers, whereas the later means that the matrix Q is PSD. Can you think of a PSD matrix that contains both negative and nonnegative elements?

In this exercise, there are two tasks for you:

1. Show that if P is column-stochastic then $(P^T x)_i \leq x_{max}$

Solution:

2. Assume P is column-stochastic. Show using LP duality that $\exists y \in \mathbb{R}^n$ such that $Py = y$, $y \geq 0$ and $\mathbf{1}^T y = 1$.

Solution:

Exercise 7

Consider the LP

$$\begin{array}{ll}\min_x & -2x_1 - 2x_2 \\ s.t. & x_1 - x_2 \leq 2 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0\end{array}$$

1. Convert the problem into standard form, construct a BFS where $(x_1, x_2) = (0, 0)$

Solution:

2. Carry out 1 iteration of the Simplex method

Solution: